

Additional HVarPart sensitivity analysis

Additional analysis and correction

During additional quality-control checks of the hierarchical variation partitioning results, we identified an error in our previous post-processing: negative adjusted contributions were not retained. This issue was identified by the authors rather than raised by the reviewers. We replaced the previous pathway with a single extractor that preserves HVarPart's `Unique`, `Average.share`, `Individual`, `I.perc(%)`, and total adjusted R^2 values.

We compared three reporting profiles:

1. **Signed:** retain each unmodified individual contribution.
2. **Zero-truncated:** replace negative individual contributions with exactly zero and recompute the denominator from the truncated contributions.
3. **Exclude-negative:** exclude the complete model if either predictor has a negative individual contribution.

Within reporting group g , the signed allocation of predictor p is

$$A_{p,g} = \frac{\sum_m I_{p,m}}{\sum_m R_{\text{adj},m}^2},$$

where $I_{p,m}$ is HVarPart's unmodified individual contribution. Models are otherwise excluded only when required predictors are absent, the denominator is non-finite or non-positive, or an individual contribution is non-finite. Every exclusion is recorded in `Outputs/Tables/HVarPart/hvarpart_model_audit.csv`.

A negative signed allocation is an adjusted partitioning result. It is not evidence of a negative ecological effect.

Results and presentation

For spatial SPD models, the zero-truncated allocations underlying the main balance figure are 75.6% for climate and 24.4% for human impact. The corresponding unmodified signed allocations are 76.4% and 23.6%. Excluding complete models with any negative contribution gives 72.4% for climate.

Across all five analyses, climate remains larger than human impact under all three profiles: yes.

Because the profiles support the same overall interpretation, the main figures use the bounded difference between paired zero-truncated human and climate allocations. Figure 2 applies this balance spatially, Figure 3 through time, and Figure 4 within its importance insets. Signed versions of Figures 2–4 are provided in the Supplementary Information, and the exclude-negative analysis is reported as an additional sensitivity profile.

Table 1: Model-level HVarPart eligibility audit. Counts are not duplicated across predictor rows.

Analysis	Models	Eligible	Excluded	Neg. unique	Neg. individual
h2 spd	31	31	0	1	0
spatial events	1262	257	1005	75	34
spatial spd	1262	1118	144	481	265
temporal events	85	76	9	13	10
temporal spd	70	69	1	14	6

Figure changes

Figure 2 now expresses the complementary human and climate allocations as human minus climate importance. The bounded scale runs from -1 (climate impact) through 0 (equal importance) to 1 (human impact). For each continent, the figure combines the continent-wide distribution of record-level balances, pooled climate-zone balances, and a horizontal line for the pooled continental balance. Densities use a continuous climate-to-neutral-to-human colour scale; the climate-zone panels use the same scale as a pale background while their summary points retain categorical climate-zone colours. Maps and continental density panels do not use the background gradient. The supplementary counterpart reports the unmodified signed human allocation on a focused -0.5 to 1.5 display while retaining every value in its source table.

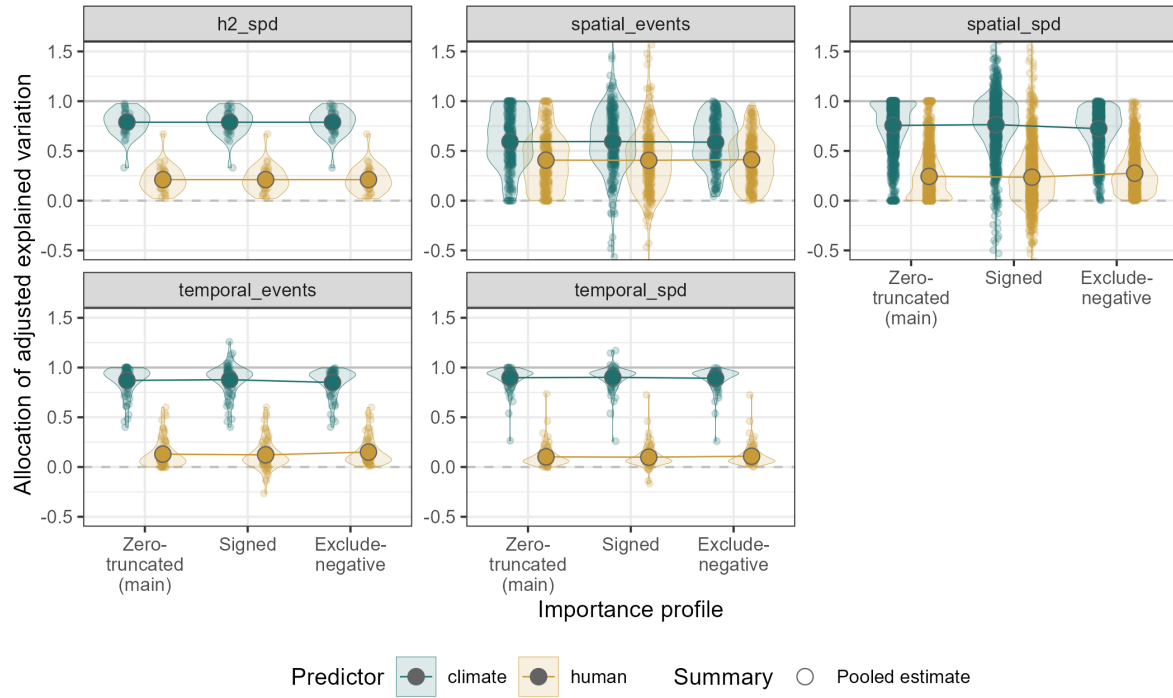
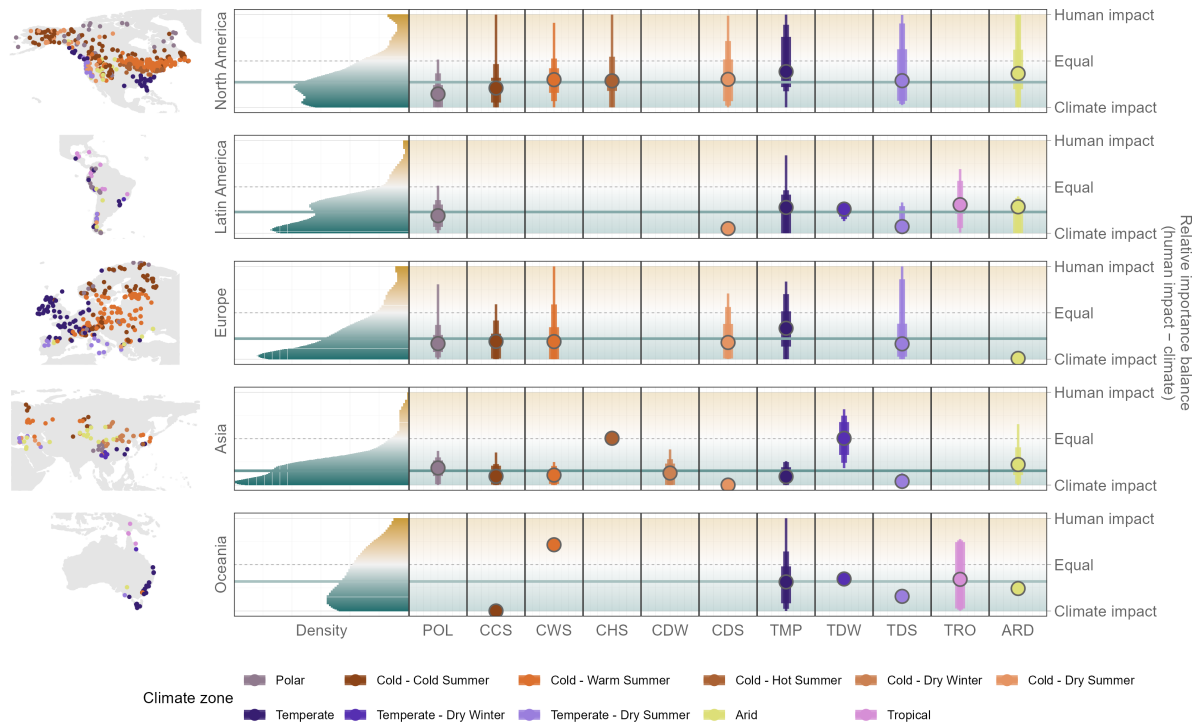
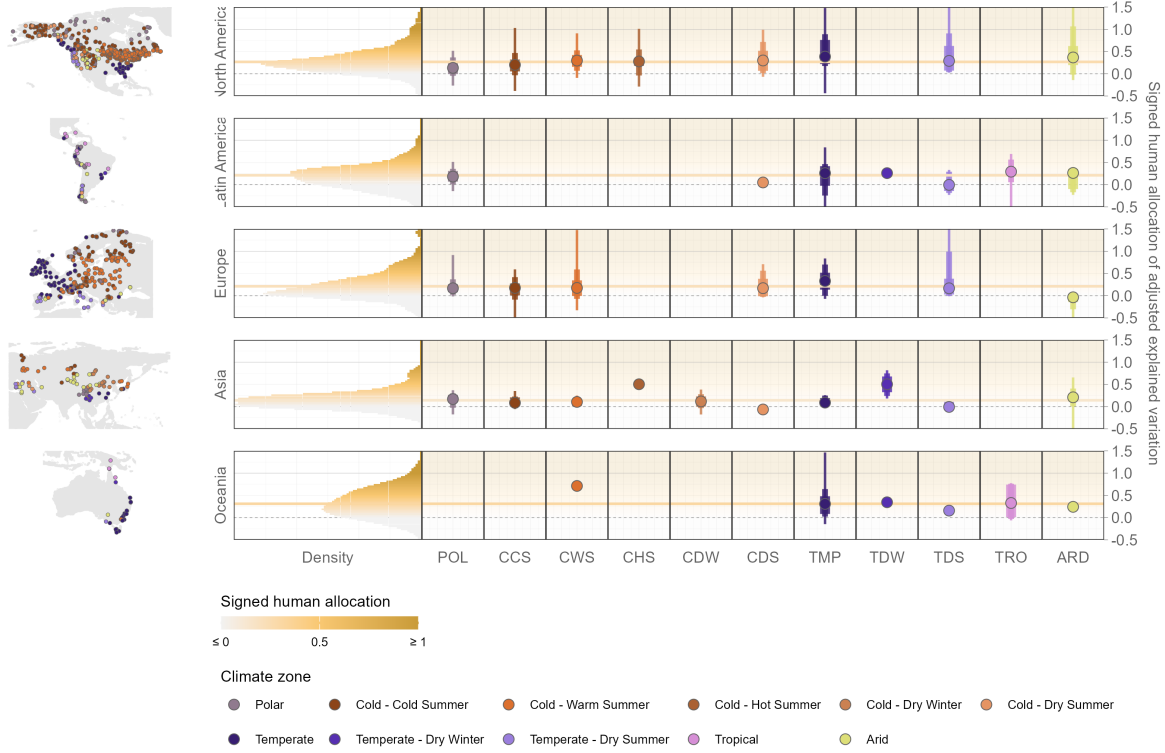


Figure 1: Zero-truncated main-figure allocation and two sensitivity profiles, calculated from the preserved fitted HVarPart results.

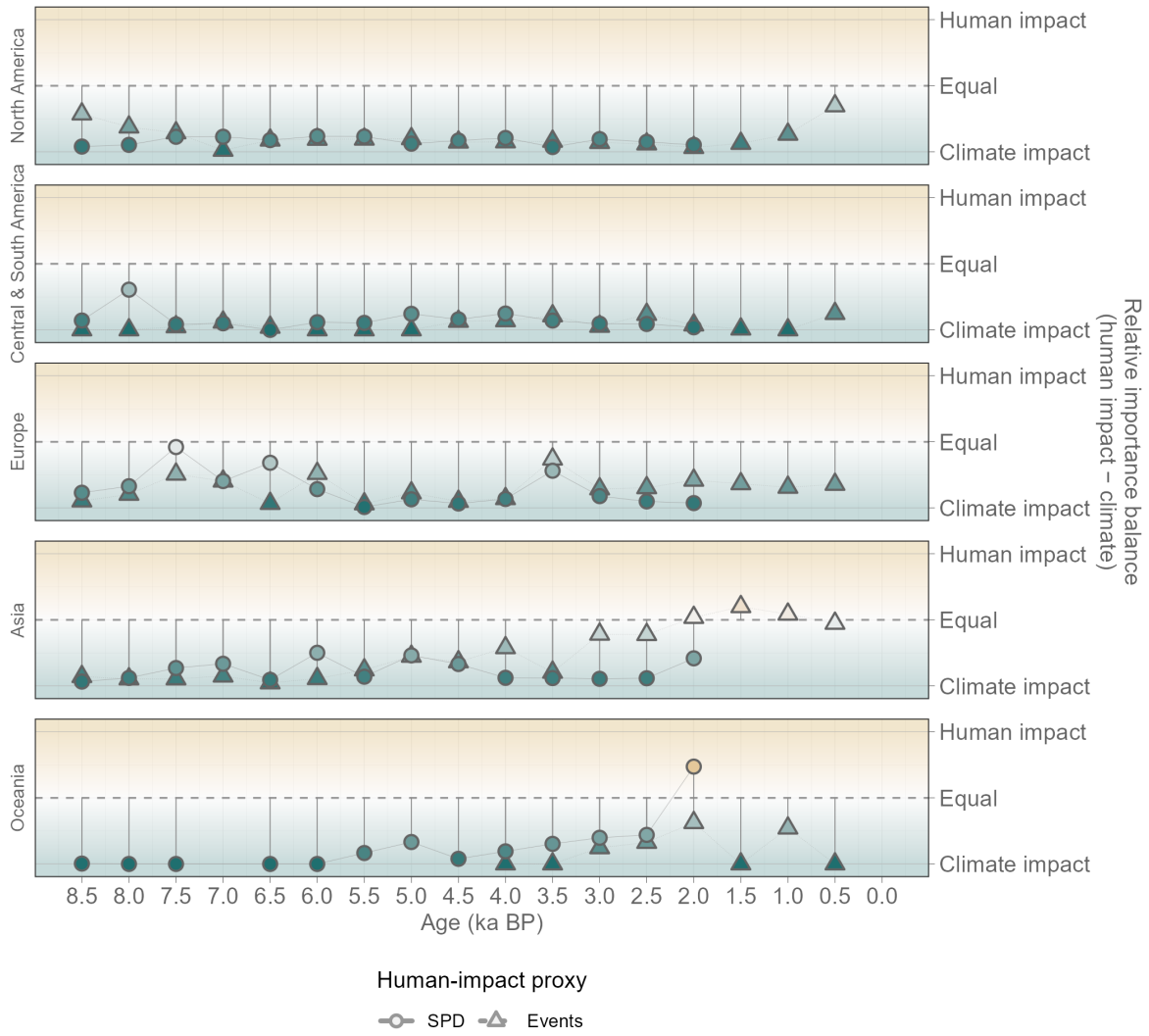


Revised main manuscript Figure 2. Spatial balance between zero-truncated human and climate allocations.

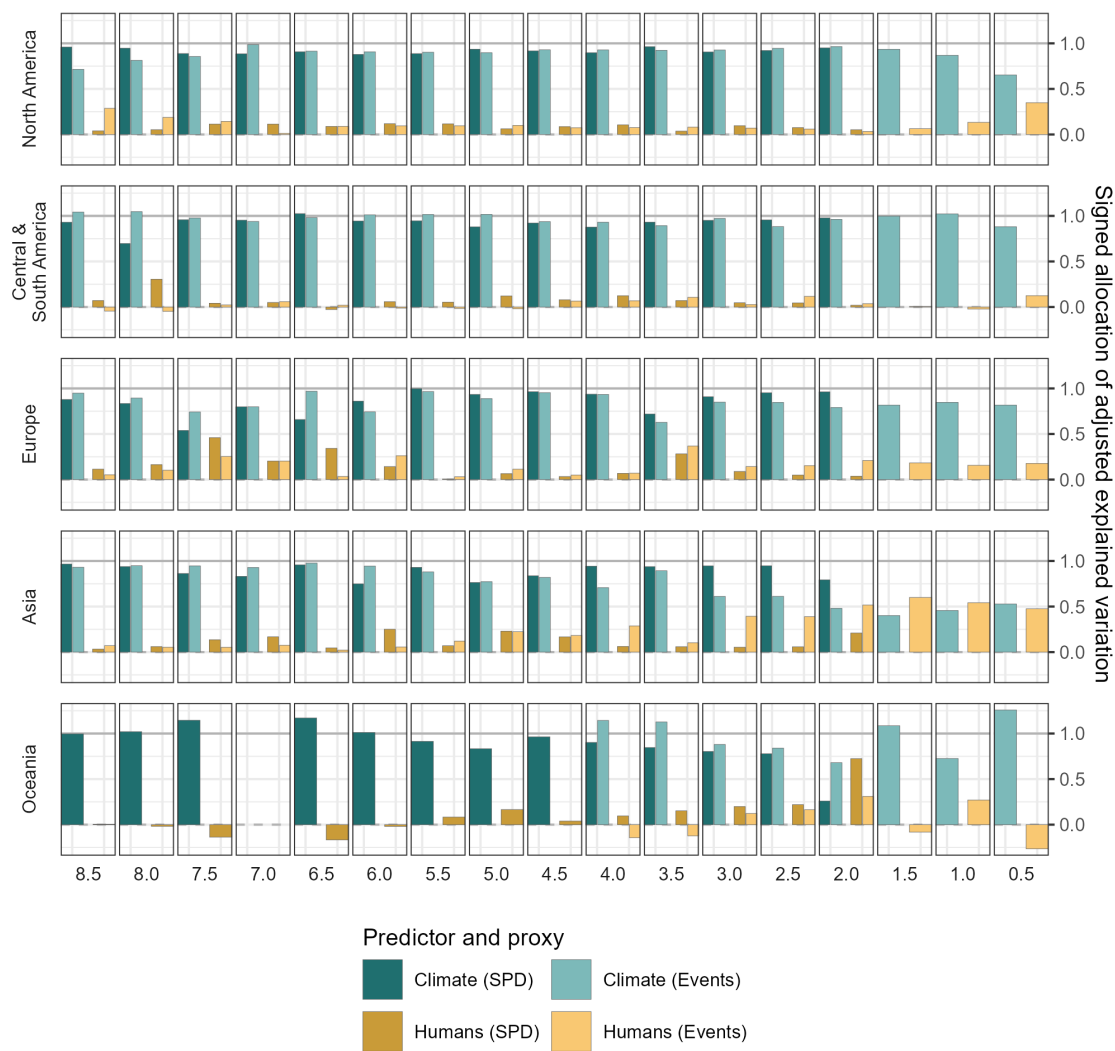


Supplementary signed Figure 2. Unmodified signed human allocation, displayed from -0.5 to 1.5; all values are retained in the source table.

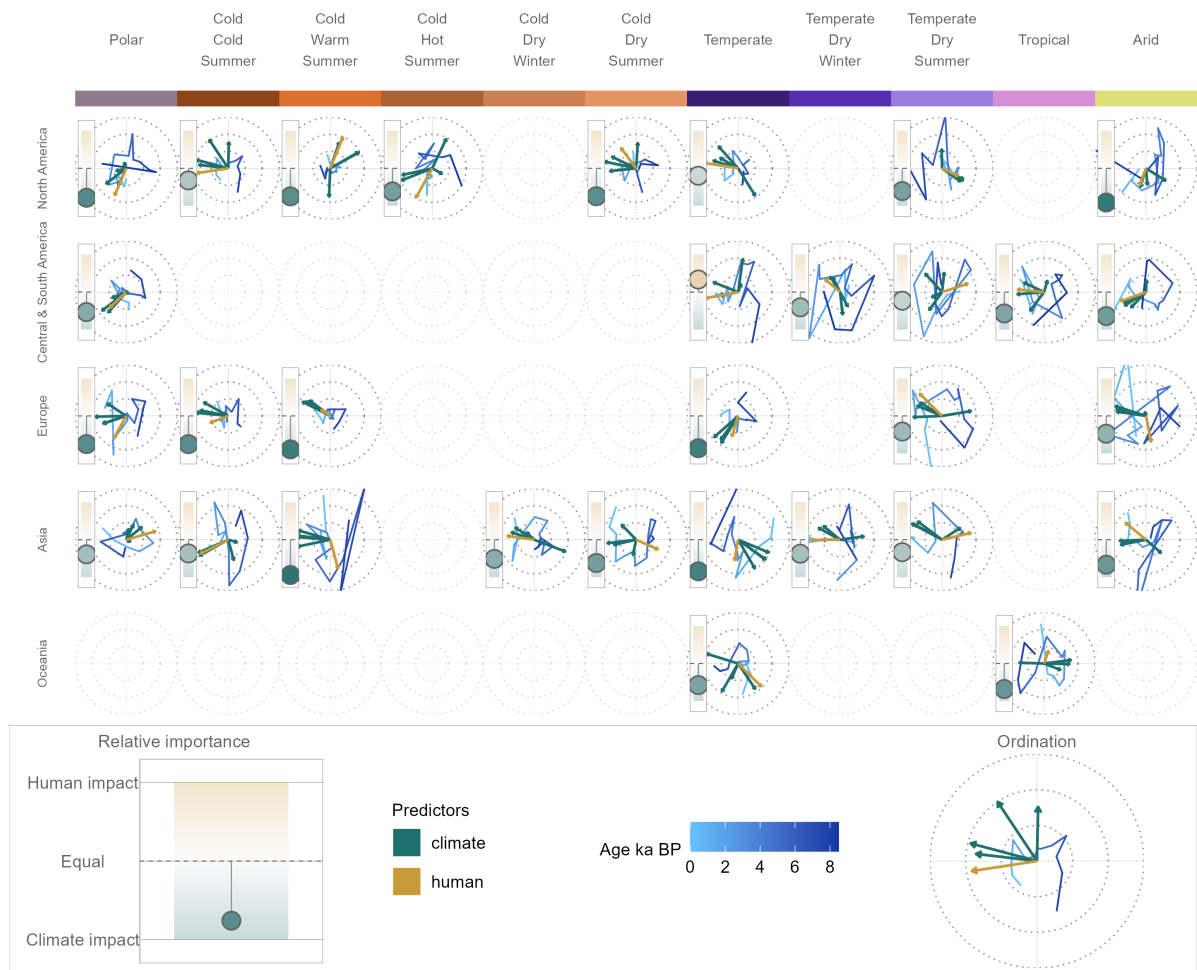
The main temporal H1 figure shows the balance through time as lollipops drawn from equal importance to each estimate. Point shape distinguishes the SPD and event human-impact proxies, while point colour uses the same climate-neutral-human scale as Figure 2 against a pale version of that gradient. SPD estimates are restricted to 2–8.5 ka BP, whereas event-based estimates span 0.5–8.5 ka BP. Its supplementary counterpart applies the same age domains and shows the unmodified signed allocations as grouped, unstacked bars with zero and one reference lines. The H2 main-figure insets likewise use one outlined balance lollipop per region and climate zone against the pale gradient; supplementary insets retain separate unmodified signed human and climate allocations.



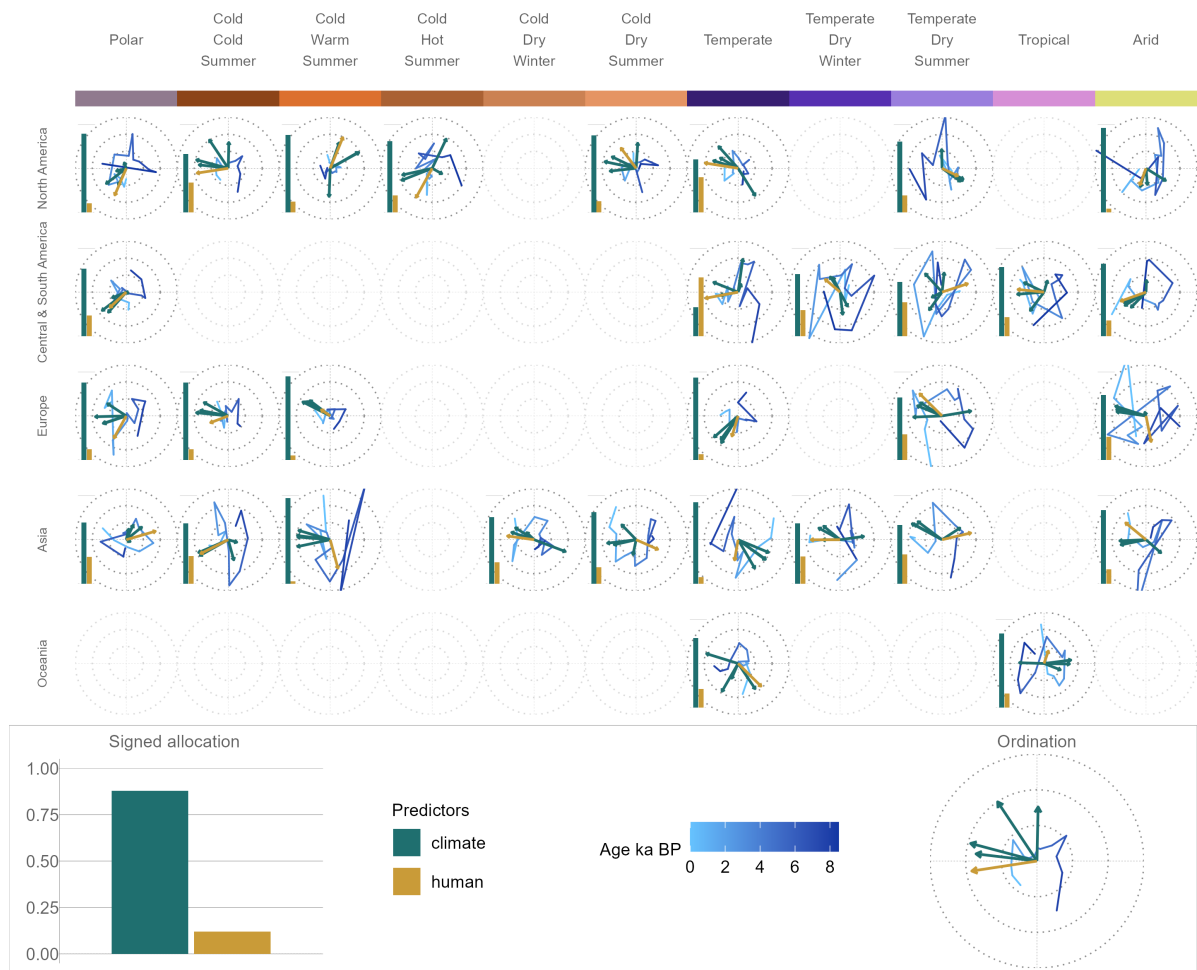
Revised main manuscript Figure 3. Temporal balance between zero-truncated human and climate allocations.



Supplementary signed Figure 3. Unmodified signed climate and human allocations through time.



Revised main manuscript Figure 4. Multidimensional community-change ordinations with balance lollipops for human and climate importance.



Supplementary signed Figure 4. Multidimensional community-change ordinations with separate unmodified signed human and climate allocations.